

Recall:

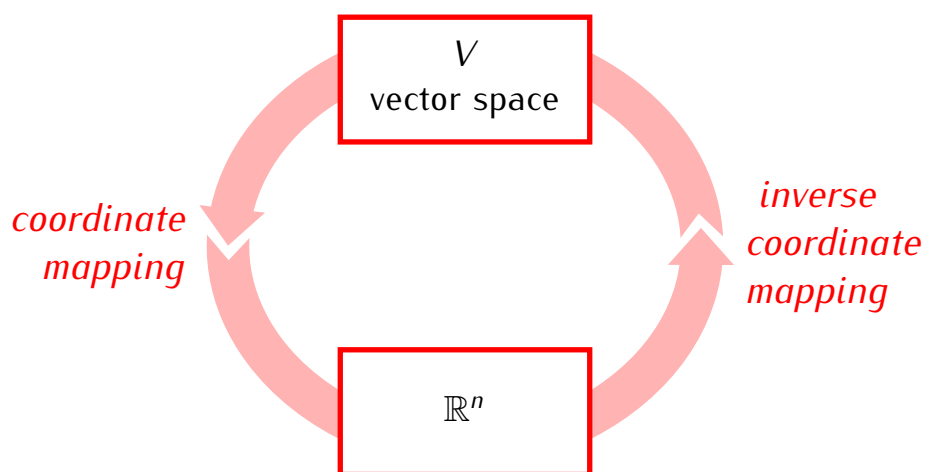
- A vector space is a set V equipped with operations of addition and multiplication by scalars. These operations must satisfy some properties.
- Some examples of vector spaces:
 - 1) $\mathbb{R}^n$ = the vector space of column vectors.
 - 2) $\mathcal{F}(\mathbb{R})$ = the vector space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$.
 - 3) $C(\mathbb{R})$ = the vector space of all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$.
 - 4) $C^\infty(\mathbb{R})$ = the vector space of all smooth functions $f: \mathbb{R} \rightarrow \mathbb{R}$.
 - 5) $M_{m,n}(\mathbb{R})$ = the vector space of all $m \times n$ matrices.
 - 6) $\mathbb{P}$ = the vector space of all polynomials.
 - 7) $\mathbb{P}_n$ = the vector space of polynomials of degree $\leq n$.
- If V, W are vector spaces then a linear transformation is a function $T: V \rightarrow W$ such that
 - 1) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
 - 2) $T(c\mathbf{v}) = cT(\mathbf{v})$
- Many problems involving $\mathbb{R}^n$ can be easily solved using row reduction, matrix multiplication etc.
- The same types of problems involving other vector spaces can be difficult to solve.

Next goal:

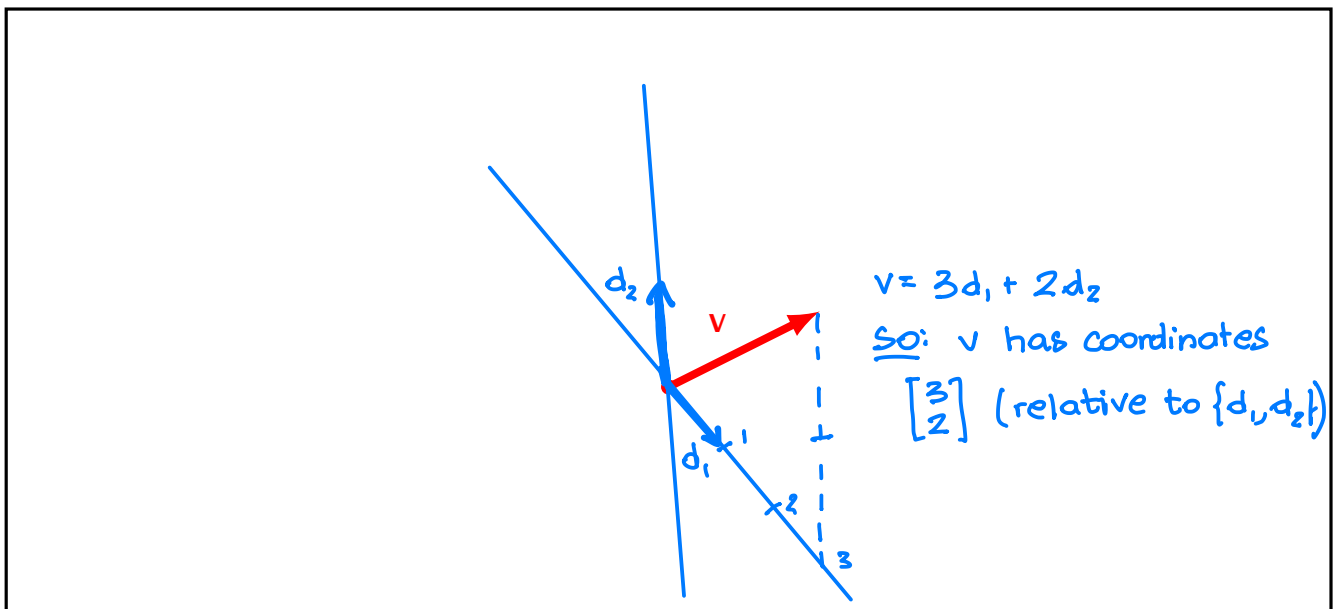
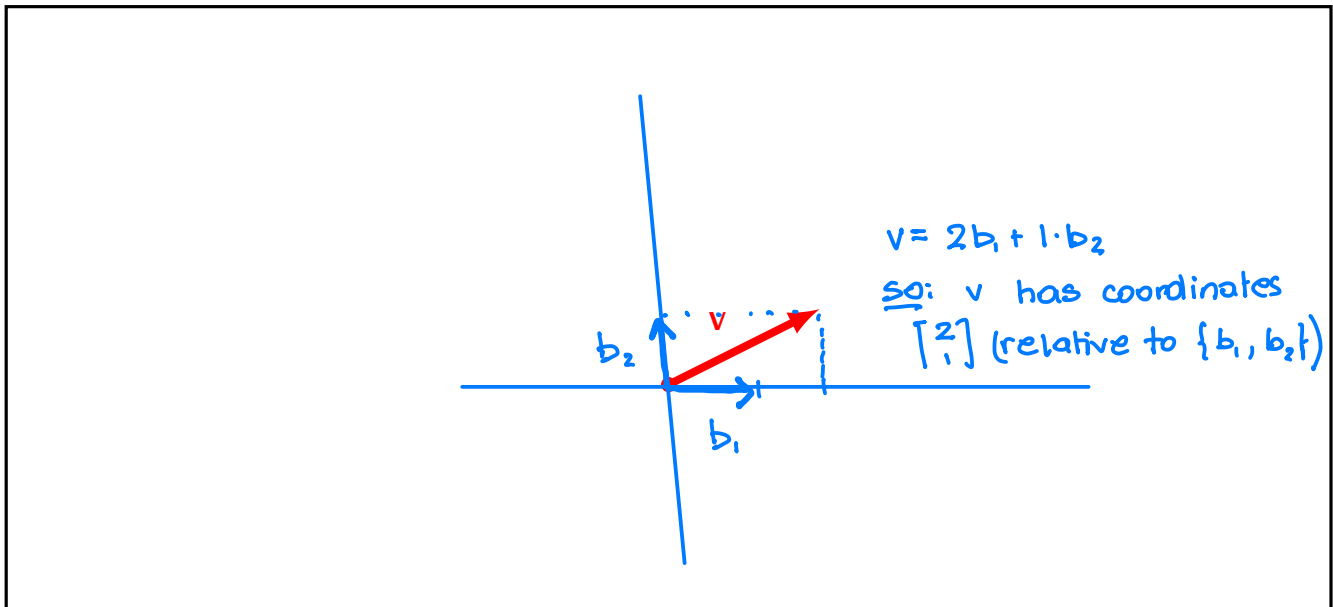
If V is a *finite dimensional* vector space then we can construct a *coordinate mapping*

$$V \rightarrow \mathbb{R}^n$$

which lets us turn computations in V into computations in $\mathbb{R}^n$.



Motivation: How to assign coordinates to vectors



Upshot: In order to define a coordinate system in a vector space V , we need to select vectors $b_1, \dots, b_p$ such that any vector v can be written as

$$v = c_1 b_1 + c_2 b_2 + \dots + c_p b_p$$

in a unique way. Then v will have coordinates $\begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$ relative to $\{b_1, b_2, \dots, b_p\}$.

Definition

If V is a vector space then vector $w \in V$ is a *linear combination* of vectors $v_1, \dots, v_p \in V$ if there exist scalars $c_1, \dots, c_p$ such that

$$w = c_1 v_1 + \dots + c_p v_p$$

Definition

If V is a vector space and $v_1, \dots, v_p$ are vectors in V then

$$\text{Span}(v_1, \dots, v_p) = \left\{ \begin{array}{l} \text{the set of all} \\ \text{linear combinations} \\ c_1 v_1 + \dots + c_p v_p \end{array} \right\}$$

Note: If $v_1, \dots, v_p \in V$ are vectors such that

$$\text{Span}(v_1, \dots, v_p) = V$$

then every vector $w \in V$ can be written as

$$w = c_1 v_1 + c_2 v_2 + \dots + c_p v_p$$

for some $c_1, \dots, c_p \in \mathbb{R}$.

Definition

If V is a vector space and $v_1, \dots, v_p \in V$ then the set $\{v_1, \dots, v_p\}$ is *linearly independent* if the homogenous equation

$$x_1 v_1 + \dots + x_p v_p = \mathbf{0}$$

has only one, trivial solution $x_1 = 0, \dots, x_p = 0$. Otherwise the set is *linearly dependent*.

Theorem

Let V be a vector space, and let $v_1, \dots, v_p \in V$. If the set $\{v_1, \dots, v_p\}$ is linearly independent then the equation

$$x_1 v_1 + \dots + x_p v_p = \mathbf{w}$$

has exactly one solution for any vector $\mathbf{w} \in \text{Span}(v_1, \dots, v_p)$.

Proof: The same as for vector equations in $\mathbb{R}^n$.

Definition

A *basis* of a vector space V is an ordered set of vectors

$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$$

such that

- 1) $\text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_n) = V$
- 2) The set $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is linearly independent.

Example:

In $\mathbb{R}^n$ let $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$, ..., $\mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$

The set $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$.

This basis is called the standard basis of $\mathbb{R}^n$

Example:

In $\mathbb{R}^2$ take $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The set $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis of $\mathbb{R}^2$.

Check: 1) the set $\{\mathbf{b}_1, \mathbf{b}_2\}$ is linearly independent

2) $\text{Span}(\mathbf{b}_1, \mathbf{b}_2) = \mathbb{R}^2$, since if $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \in \mathbb{R}^2$ then

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (a_2 - a_1) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = a_1 \mathbf{b}_1 + (a_2 - a_1) \mathbf{b}_2$$

Example:

Let $\mathbb{P}_n$ = the vector space of polynomials of degree $\leq n$

$$= \{a_0 + a_1 t + \dots + a_n t^n \mid a_i \in \mathbb{R}\}$$

The set $\mathcal{E} = \{1, t, t^2, \dots, t^n\}$ is a basis of $\mathbb{P}_n$.

This basis is called the standard basis of $\mathbb{P}_n$.

Theorem

A set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis of a vector space V if and only if for each $\mathbf{v} \in V$ the vector equation

$$x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n = \mathbf{v}$$

has a unique solution.

Proof: Since $\text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_n) = V$ thus for each $\mathbf{v} \in V$ the equation

$$x_1 \mathbf{b}_1 + \dots + x_n \mathbf{b}_n = \mathbf{v}$$

has a solution.

Since the set $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is linearly independent, this solution is unique.

Definition

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of a vector space V . For $\mathbf{v} \in V$ let $c_1, \dots, c_n$ be the unique numbers such that

$$c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n = \mathbf{v}$$

Then the vector

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

is called the *coordinate vector of $\mathbf{v}$ relative to the basis $\mathcal{B}$* and it is denoted by $[\mathbf{v}]_{\mathcal{B}}$.

Example. Let $\mathcal{E} = \{1, t, t^2\}$ be the standard basis of $\mathbb{P}_2$, and let

$$p(t) = 3 + 2t - 4t^2$$

Find the coordinate vector $[p]_{\mathcal{E}}$.

Solution: We have:

$$p(t) = 3 \cdot \underbrace{1}_{\substack{\uparrow \\ \text{1st vector} \\ \text{of } \mathcal{E}}} + 2 \cdot \underbrace{t}_{\substack{\uparrow \\ \text{2nd vector} \\ \text{of } \mathcal{E}}} + (-4) \cdot \underbrace{t^2}_{\substack{\uparrow \\ \text{3rd vector} \\ \text{of } \mathcal{E}}}$$

so:

$$[p(t)]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}$$

Example. Let $\mathcal{B} = \{1, 1 + t, 1 + t + t^2\}$. One can check that $\mathcal{B}$ is a basis of $\mathbb{P}_2$.
Let

$$p(t) = 3 + 2t - 4t^2$$

Find the coordinate vector $[p]_{\mathcal{B}}$.

Solution: We have:

$$p(t) = 1 \cdot \underbrace{1}_{\substack{\uparrow \\ \text{1st vector} \\ \text{of } \mathcal{B}}} + 6 \cdot \underbrace{(1+t)}_{\substack{\uparrow \\ \text{2nd vector} \\ \text{of } \mathcal{B}}} + (-4) \cdot \underbrace{(1+t+t^2)}_{\substack{\uparrow \\ \text{3rd vector} \\ \text{of } \mathcal{B}}}$$

so:

$$[p(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 6 \\ -4 \end{bmatrix}$$

Example. Consider the following vectors in $\mathbb{R}^2$:

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{b}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

One can check that $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis of $\mathbb{R}^2$. Find the coordinate vector $[\mathbf{v}]_{\mathcal{B}}$.

Solution:

We have $[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ where $c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 = \mathbf{v}$.

Thus it suffices to solve the equation $x_1 \mathbf{b}_1 + x_2 \mathbf{b}_2 = \mathbf{v}$.

Augmented matrix :

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 1 & -1 & 0 \end{array} \right] \xrightarrow{\text{row reduction}} \left[\begin{array}{cc|c} 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{array} \right]$$

$$\text{Thus: } \begin{cases} x_1 = 1/3 \\ x_2 = 1/3 \end{cases}$$

so:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$$

Proposition

If $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis of a vector space V then

$$1) [\mathbf{v} + \mathbf{w}]_{\mathcal{B}} = [\mathbf{v}]_{\mathcal{B}} + [\mathbf{w}]_{\mathcal{B}}.$$

$$2) [c\mathbf{v}]_{\mathcal{B}} = c[\mathbf{v}]_{\mathcal{B}}$$

for any $\mathbf{v}, \mathbf{w} \in V, c \in \mathbb{R}$.